

Screenshots CubeSat Telemetry Anomaly Detection Assistant App

Analyze telemetry page

Analysis source

Data profile

Authentic OPS-SAT-AD v2

Review recorded data on this computer.

The app does not upload or download data.

Analyze telemetry

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Analyze telemetry

Data and AI walkthrough

Analyze telemetry

This app helps you decide which recorded spacecraft measurements need a closer look. Each result describes a segment: a group of readings from one telemetry channel over a time window. Start with the counts below, then choose a segment to inspect its results.

Artificial intelligence (AI) supplies an anomaly probability, a review status, and similar historical examples. You decide whether more review is needed. These results do not confirm a spacecraft fault.

Active data profile: Authentic OPS-SAT-AD v2

Historical OPS-SAT Anomaly Detection (OPS-SAT-AD) v2 data from Zenodo record 15108715

Each segment ID identifies one example. Channel codes group examples by telemetry source; they do not identify separate spacecraft or count events within a segment.

Held-out means these examples were kept out of model training and threshold selection. They let us check the model on separate data.

Understand the data

The active Authentic OPS-SAT-AD v2 contains 2,123 segments, 18 numerical features per segment, and 9 coded channels. Its 1,594 official training-role segments become 1,275 model-training rows plus 319 threshold-validation rows. The remaining 529 segments are held out.

Row role	Segments	Purpose
Model training	1275	Learn the feature scales and neural-network weights.
Threshold validation	319	Choose the classification threshold after model fitting.
Held-out test	529	Screen segments and evaluate predictions after fitting.

The held-out population is the set being screened, not a count of anomalies. Reviewed normal/anomalous labels are revealed only for evaluation; AI screening statuses come from the current model run.

Terms used on this page

- Telemetry:** Measurements recorded by a spacecraft system over time.
- Dataset:** An organized collection of records; this project reads local comma-separated value (CSV) files.
- Segment:** One time-window summary and the unit screened by the model.
- Channel:** A coded source identifier supplied by OPS-SAT-AD, not a subsystem name here.
- Numerical feature:** One measured summary value used as an input to the model.
- Reviewed label:** The historical normal or anomalous class recorded for evaluation and examples.
- Anomaly probability:** The neural network's 0-to-1 estimate for one segment, not certainty.
- Threshold:** The validation-selected probability boundary used for a binary prediction.
- Uncertainty band:** The plus-or-minus margin around the threshold that routes close values to review.
- AI screening status:** The mutually exclusive flagged, uncertain, or likely-normal result for one segment.

Held-out screening summary

Each held-out segment receives exactly one status. The three status counts add up to the screened total.

Segments screened

529

Flagged anomaly

104

Uncertain review

3

Likely normal

422

Flagged anomaly means the probability is above the uncertainty band; Uncertain review means it is inside the band; Likely normal means it is below the band.

These screening counts summarize model decisions. Evaluation metrics compare predictions with reviewed labels after prediction. Software tests separately check that the code and interface behave as specified; they are not model-performance scores.

This retrospective assistant provides AI screening evidence for human review. It does not confirm a fault, control a spacecraft, or replace human judgment.

Channel comparison

The horizontal labels are coded source identifiers, not inferred subsystem names. The vertical axis is held-out segment count. Bars compare flagged and uncertain counts in ascending channel-code order; printed values and the table preserve meaning without color or pointer hover.

Segments needing review by telemetry channel

Telemetry channel	Flagged anomaly	Uncertain review
CADC0872	29	0
CADC0873	30	0
CADC0874	20	1
CADC0884	0	0
CADC0886	1	0
CADC0888	11	0
CADC0890	2	0
CADC0892	6	0
CADC0894	5	2

Channel counts and rates

Each row uses one channel's screened count as its denominator. Count ranking answers which channel has the most segments; rate ranking answers which has the largest share, so the two rankings can differ.

Channel code	Screened	Flagged anomaly	Uncertain review	Likely normal	Flagged rate	Uncertain rate
CADC0872	132	29	0	103	21.97%	0%
CADC0873	153	30	0	123	19.61%	0%
CADC0874	52	20	1	31	38.46%	1.92%
CADC0884	36	0	0	36	0%	0%
CADC0886	4	1	0	3	25%	0%
CADC0888	64	11	0	53	17.19%	0%
CADC0890	2	2	0	0	100%	0%
CADC0892	53	6	0	47	11.32%	0%
CADC0894	33	5	2	26	15.15%	6.06%

- **CADC0872** — 29 flagged (22.0%), 0 uncertain (0.0%), and 103 likely normal of 132 screened.
- **CADC0873** — 30 flagged (19.6%), 0 uncertain (0.0%), and 123 likely normal of 153 screened.
- **CADC0874** — 20 flagged (38.5%), 1 uncertain (1.9%), and 31 likely normal of 52 screened.
- **CADC0884** — 0 flagged (0.0%), 0 uncertain (0.0%), and 36 likely normal of 36 screened.
- **CADC0886** — 1 flagged (25.0%), 0 uncertain (0.0%), and 3 likely normal of 4 screened.
- **CADC0888** — 11 flagged (17.2%), 0 uncertain (0.0%), and 53 likely normal of 64 screened.
- **CADC0890** — 2 flagged (100.0%), 0 uncertain (0.0%), and 0 likely normal of 2 screened.
- **CADC0892** — 6 flagged (11.3%), 0 uncertain (0.0%), and 47 likely normal of 53 screened.
- **CADC0894** — 5 flagged (15.2%), 2 uncertain (6.1%), and 26 likely normal of 33 screened.

Worked example for CADC0873: flagged rate = 30 flagged / 153 screened = 19.6%; uncertain rate = 0 uncertain / 153 screened = 0.0%.

Highest flagged count: CADC0873 (30 held-out segments).

Ranked segment review – All

Ranked segment review

The highest anomaly probabilities appear first. Equal scores are ordered by segment ID. Filters narrow this list without retraining the model or changing the overall counts.

Leave status or channel empty to include all available values.

AI screening status

Telemetry channel

Exact segment ID

All statuses

All channels

For example, demo-046

Reset filters

Showing 529 of 529 held-out segments.

Segment ID	Channel	AI status	Probability
1068	CADC0888	Flagged anomaly	100%
444	CADC0873	Flagged anomaly	100%
1189	CADC0873	Flagged anomaly	100%
1725	CADC0890	Flagged anomaly	100%
243	CADC0874	Flagged anomaly	100%
1183	CADC0873	Flagged anomaly	100%
320	CADC0874	Flagged anomaly	100%
235	CADC0892	Flagged anomaly	100%
309	CADC0874	Flagged anomaly	100%
239	CADC0874	Flagged anomaly	100%

Segment to inspect

Select an exact segment ID

Select a segment ID to inspect its probability, threshold, recommended next steps, deviations, and similar reviewed cases.

Ranked segment review – Falg anomaly

Ranked segment review

The highest anomaly probabilities appear first. Equal scores are ordered by segment ID. Filters narrow this list without retraining the model or changing the overall counts.

Leave status or channel empty to include all available values.

AI screening status

Flagged anomaly

Telemetry channel

All channels

Exact segment ID

For example, demo-046

Reset filters

Showing 104 of 529 held-out segments.

Segment ID	Channel	AI status	Probability
1068	CADC0888	Flagged anomaly	100%
444	CADC0873	Flagged anomaly	100%
1189	CADC0873	Flagged anomaly	100%
1725	CADC0890	Flagged anomaly	100%
243	CADC0874	Flagged anomaly	100%
1183	CADC0873	Flagged anomaly	100%
320	CADC0874	Flagged anomaly	100%
235	CADC0892	Flagged anomaly	100%
309	CADC0874	Flagged anomaly	100%
239	CADC0874	Flagged anomaly	100%

Segment to inspect

1068

Selected segment: 1068

Telemetry channel CADC0888 · held-out test segment · Flagged anomaly

Anomaly probability

100.0%

Validation threshold

71.0%

Uncertainty margin

±5.0%

AI screening status

Flagged anomaly

Anomaly probability is the neural network's 0-to-1 estimate, not certainty. The validation threshold is the cutoff for predicting normal or anomalous. The uncertainty margin adds a band around that cutoff for close calls needing review. Percentages are rounded to one decimal place, so even 100.0% does not guarantee a correct prediction.

The probability is above the 76.0% upper edge of the uncertainty band, so the AI flags this segment for review.

Recommended next steps

For segment 1068, channel CADC0888. These are proposed human reviews. The actions are simulated; the app does not perform them, inspect spacecraft hardware, or confirm that a review occurred.

Review the evidence in this order using the results above and below:

1. Review the model probability and validation-selected threshold.
2. Inspect class-aware normal and anomalous historical cases.

Total simulated review effort: 2 units. These coursework costs represent relative effort, not minutes or money. A-star planning orders the required checks according to their prerequisites and costs.

To see all five available checks and their prerequisites, open Data and AI walkthrough and select A-star planning.

Largest feature deviations

These features differ most from the reviewed normal model-training examples. The program compares each value with that group's average and spread. This comparison helps locate unusual measurements; it does not explain their physical cause. The MLP probability stage in Data and AI walkthrough explains all 18 feature names.

Rank	Feature
1	diff2_var
2	diff_var
3	kurtosis

Ranked segment review – Uncertain review

Ranked segment review

The highest anomaly probabilities appear first. Equal scores are ordered by segment ID. Filters narrow this list without retraining the model or changing the overall counts.

Leave status or channel empty to include all available values.

AI screening status

Uncertain review

Telemetry channel

All channels

Exact segment ID

For example, demo-046

Reset filters

Showing 3 of 529 held-out segments.

Segment ID	Channel	AI status	Probability
1756	CADC0894	Uncertain review	73.19%
278	CADC0874	Uncertain review	70.39%
1771	CADC0894	Uncertain review	66.56%

Segment to inspect

1756

Selected segment: 1756

Telemetry channel CADC0894 · held-out test segment · Uncertain review

Anomaly probability

73.2%

Validation threshold

71.0%

Uncertainty margin

±5.0%

AI screening status

Uncertain review

Anomaly probability is the neural network's 0-to-1 estimate, not certainty. The validation threshold is the cutoff for predicting normal or anomalous. The uncertainty margin adds a band around that cutoff for close calls needing review. Percentages are rounded to one decimal place, so even 100.0% does not guarantee a correct prediction.

The probability is inside the uncertainty band from 66.0% to 76.0%, so this segment is routed to human review.

Review context: probability within the uncertainty band

Recommended next steps

For segment 1756, channel CADC0894. These are proposed human reviews. The actions are simulated; the app does not perform them, inspect spacecraft hardware, or confirm that a review occurred.

Review the evidence in this order using the results above and below:

- 1. Review the model probability and validation-selected threshold.
- 2. Inspect class-aware normal and anomalous historical cases.
- 3. Escalate the evidence package for final human interpretation.

Total simulated review effort: 3 units. These coursework costs represent relative effort, not minutes or money. A-star planning orders the required checks according to their prerequisites and costs.

To see all five available checks and their prerequisites, open Data and AI walkthrough and select A-star planning.

Largest feature deviations

These features differ most from the reviewed normal model-training examples. The program compares each value with that group's average and spread. This comparison helps locate unusual measurements; it does not explain their physical cause. The MLP probability stage in Data and AI walkthrough explains all 18 feature names.

Rank	Feature
1	kurtosis
2	n_peaks
3	skew

Similar reviewed historical cases

Distance compares two segments across the same 18 scaled features; smaller values mean more similar examples. The normal and anomalous lists are searched separately and can include other channels. Reviewed labels describe those historical cases; they do not change this segment's prediction. The largest feature gaps show where two examples differ, not which features caused the MLP prediction.

Reviewed normal cases

Rank 1: segment 1749

Reviewed label: normal · Euclidean distance: 2.697

Largest standardized feature gaps: smooth10_n_peaks (2.686), skew (0.212), kurtosis (0.085)

Rank 2: segment 1750

Reviewed label: normal · Euclidean distance: 4.232

Largest standardized feature gaps: n_peaks (3.605), kurtosis (1.911), skew (1.069)

Rank 3: segment 1754

Reviewed label: normal · Euclidean distance: 5.035

Largest standardized feature gaps: kurtosis (3.022), skew (2.465), diff2_peaks (1.869)

Reviewed anomalous cases

Rank 1: segment 1772

Reviewed label: anomalous · Euclidean distance: 4.466

Largest standardized feature gaps: n_peaks (4.056), len (0.927), len_weighted (0.805)

Rank 2: segment 875

Reviewed label: anomalous · Euclidean distance: 4.743

Largest standardized feature gaps: n_peaks (4.056), kurtosis (1.498), len (1.094)

Rank 3: segment 907

Reviewed label: anomalous · Euclidean distance: 5.339

Largest standardized feature gaps: kurtosis (3.296), skew (3.144), n_peaks (2.253)

Historical evidence summary: normal.



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Ranked segment review – Likely normal

Ranked segment review

The highest anomaly probabilities appear first. Equal scores are ordered by segment ID. Filters narrow this list without retraining the model or changing the overall counts.

Leave status or channel empty to include all available values.

AI screening status

Likely normal x

Telemetry channel

All channels

Exact segment ID

For example, demo-046

Reset filters

Showing 422 of 529 held-out segments.

Segment ID	Channel	AI status	Probability
1955	CADC0872	Likely normal	58.87%
564	CADC0873	Likely normal	56.34%
256	CADC0874	Likely normal	49.94%
1956	CADC0872	Likely normal	49.6%
458	CADC0873	Likely normal	48.09%
132	CADC0872	Likely normal	45.52%
42	CADC0872	Likely normal	39.52%
452	CADC0873	Likely normal	37.49%
1190	CADC0873	Likely normal	32.53%
1735	CADC0894	Likely normal	32.27%

Segment to inspect

1955

Selected segment: 1955

Telemetry channel CADC0872 - held-out test segment - Likely normal

Anomaly probability

58.9%

Validation threshold

71.0%

Uncertainty margin

±5.0%

AI screening status

Likely normal

Anomaly probability is the neural network's 0-to-1 estimate, not certainty. The validation threshold is the cutoff for predicting normal or anomalous. The uncertainty margin adds a band around that cutoff for close calls needing review. Percentages are rounded to one decimal place, so even 100.0% does not guarantee a correct prediction.

The probability is below the 66.0% lower edge of the uncertainty band, so the AI marks this segment as likely normal.

Recommended next steps

For segment 1955, channel CADC0872. These are proposed human reviews. The actions are simulated; the app does not perform them, inspect spacecraft hardware, or confirm that a review occurred.

Review the evidence in this order using the results above and below:

1. Review the model probability and validation-selected threshold.
2. Inspect class-aware normal and anomalous historical cases.
3. Inspect the critical channel deviation and its contributing features.

Total simulated review effort: 4 units. These coursework costs represent relative effort, not minutes or money. A-star planning orders the required checks according to their prerequisites and costs.

To see all five available checks and their prerequisites, open Data and AI walkthrough and select A-star planning.

Largest feature deviations

These features differ most from the reviewed normal model-training examples. The program compares each value with that group's average and spread. This comparison helps locate unusual measurements; it does not explain their physical cause. The MLP probability stage in Data and AI walkthrough explains all 18 feature names.

Rank	Feature
1	gaps_squared
2	len_weighted
3	duration

Similar reviewed historical cases

Distance compares two segments across the same 18 scaled features; smaller values mean more similar examples. The normal and anomalous lists are searched separately and can include other channels. Reviewed labels describe those historical cases; they do not change this segment's prediction. The largest feature gaps show where two examples differ, not which features caused the MLP prediction.

Reviewed normal cases

Rank 1: segment 1181

Reviewed label: normal · Euclidean distance: 1.008

Largest standardized feature gaps: diff_peaks (0.476), n_peaks (0.451), diff2_peaks (0.429)

Rank 2: segment 1808

Reviewed label: normal · Euclidean distance: 1.048

Largest standardized feature gaps: diff_peaks (0.714), diff2_peaks (0.613), n_peaks (0.451)

Rank 3: segment 1943

Reviewed label: normal · Euclidean distance: 1.080

Largest standardized feature gaps: diff_peaks (0.782), diff2_peaks (0.582), n_peaks (0.451)

Reviewed anomalous cases

Rank 1: segment 1015

Reviewed label: anomalous · Euclidean distance: 0.785

Largest standardized feature gaps: n_peaks (0.451), diff2_peaks (0.429), skew (0.365)

Rank 2: segment 1817

Reviewed label: anomalous · Euclidean distance: 0.855

Largest standardized feature gaps: n_peaks (0.451), duration (0.394), len_weighted (0.391)

Rank 3: segment 1939

Reviewed label: anomalous · Euclidean distance: 1.221

Largest standardized feature gaps: duration (0.662), len_weighted (0.661), diff2_peaks (0.490)

Historical evidence summary: anomalous.



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Data and AI walkthrough page

CubeSat telemetry analysis

localhost:8501/data-and-ai-walkthrough

Ask Gemini

Analyze telemetry Data and AI walkthrough

Analysis source

Data profile ⓘ

Authentic OPS-SAT-AD v2

Review recorded data on this computer.
The app does not upload or download data.

Data and AI walkthrough

Follow one recorded telemetry segment through the artificial intelligence (AI) methods used by Analyze telemetry. Telemetry means measurements recorded by a spacecraft. A segment groups readings from one channel over a time window. The app represents it with 18 numerical features, such as its average value and the spread of its readings. An anomaly is an unusual pattern in those measurements.

The examples come from held-out test data: segments kept out of model training and threshold selection. The threshold is the probability cutoff used to classify a segment. Start with Data source and roles, then follow the stages in order or open the method you want to understand. Each stage explains its result using the selected example.

Pipeline overview

1. Check the data and separate training, validation, and test rows.
2. Scale the 18 features and estimate anomaly probability with a neural network called a multilayer perceptron (MLP).
3. Apply the cutoff and compare predictions with reviewed labels.
4. Find similar examples in the normal and anomalous groups.
5. Apply if/then rules to identify reasons for further review.
6. Use A-star to order simulated review checks.
7. See how the source code connects these methods and course concepts.

See next page

Data source and role

Choose one walkthrough stage

Data source and roles

MLP probability

Threshold and evaluation

Similarity search

Logic rules

A-star planning

Architecture map

Active data profile: Authentic OPS-SAT-AD v2. Historical OPS-SAT Anomaly Detection (OPS-SAT-AD) v2 data from Zenodo record 15108715.

Choose a segment to follow through the methods. The highest anomaly probabilities appear first. Selecting another example does not retrain or refit the model. The data overview and architecture map describe the whole program, so those views stay the same.

Example held-out test segment

1068

Selected segment 1068, channel CAD0888.

The segment ID identifies this example. The channel code identifies its telemetry source; it does not tell us which spacecraft subsystem caused an unusual reading.

Each row in dataset.csv describes one segment. The program gives each row a job: train the model, choose its cutoff, or test its predictions. Keeping the test rows separate lets us check the model on examples it has not learned from.

Input: Local OPS-SAT-AD comma-separated value (CSV) files, or the small set of simulated examples included with the app. CSV stores a table as lines of text.

Action: Check the expected file, columns, and values, then assign the row groups.

Output: A checked table of 18 features per segment, with separate row groups.

These checks prepare the model's inputs. The MLP probability stage then puts the features on a common scale and uses them to make a prediction.

A matching file can still contain imperfect historical labels. Checking the file does not establish what physically happened on the spacecraft.

Official source and local files

OPS-SAT-AD v2 is the historical anomaly-detection dataset used for the authentic profile. The source is Zenodo record 15108715. The app never downloads or replaces these files during startup or interaction.

Open Zenodo record 15108715

Official dataset.csv

Official segments.csv

How to read the file table

- File:** The official filename published with the OPS-SAT-AD v2 record.
- Local destination:** Where the user places that file for this program.
- Expected size:** The reviewed byte count; a mismatch can indicate a wrong or incomplete file.
- Expected SHA-256:** A Secure Hash Algorithm 256-bit fingerprint used to check the exact file bytes.
- Program role:** Why the program reads the file and what it may use it for.

File	Local destination	Expected size	Expected SHA-256	Program role
dataset.csv	data/dataset.csv	507,550 bytes	b524177da8f516d5c9f63c7acbc385341f0ad42046ef20c1bed2d25e51b98f02	Validated table with one row per segment and 18 numerical
segments.csv	data/segments.csv	18,987,091 bytes	ds201e9e751eb2a53a0ff7c11567dc4239f594ea4b479b2aa66fe67ddcbcb9ba	Raw 303,493-reading file used by the standalone verifier to

dataset.csv has one derived feature row per segment and is the model input. segments.csv has 303,493 raw readings and is used by the standalone verifier to cross-check segment metadata.

Optional: prepare the data files in Terminal

If authentic data is already loading, you can skip this setup. Otherwise, open Terminal in Portfolio-Project-Module-8 and run the commands below. Compare the SHA-256 results with the file table above. See data/README.md for the full setup.

```
curl -fL https://zenodo.org/api/records/15108715/files/dataset.csv/content -o data/dataset.csv
curl -fL https://zenodo.org/api/records/15108715/files/segments.csv/content -o data/segments.csv
md5 data/dataset.csv data/segments.csv
shasum -a 256 data/dataset.csv data/segments.csv
```

The app displays these commands; it does not run them.

How this dataset is divided

Authentic OPS-SAT-AD v2 has 2,123 segment rows across 9 channels. The app checks the 18 numerical features already stored for each segment. It does not calculate these summaries from raw readings during page loading.

The official raw file, segments.csv, is expected to contain 303,493 individual readings. The separate verification command checks that file and compares its segment metadata with dataset.csv.

How to read the row-role table

- Row role:** The one job assigned to that group of segment rows.
- Segments:** The number of rows in that group for the active data profile.
- Purpose:** What that group may influence in the machine-learning workflow.
- Model training:** Rows used to learn feature averages and scales, and the network's weights.
- Threshold validation:** Separate rows used after fitting to choose the probability cutoff.
- Held-out test:** Rows reserved from fitting and threshold selection for final screening and tests.

Row role	Segments	Purpose
Model training	1275	Learn the feature scales and neural-network weights.
Threshold validation	319	Choose the classification threshold after model fitting.
Held-out test	529	Screen segments and evaluate predictions after fitting.

The 1,594 official training-role rows are split into 1,275 model-training rows and 319 threshold-validation rows. The 529 held-out rows are not used to fit weights or choose the threshold. Using test answers to make those choices would be data leakage: the model would get help from the same examples later used to judge it.

MLP probability

Choose one walkthrough stage

Data source and roles	MLP probability	Threshold and evaluation	Similarity search	Logic rules	A-star planning	Architecture map
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Active data profile: Authentic OPS-SAT-AD v2. Historical OPS-SAT Anomaly Detection (OPS-SAT-AD) v2 data from Zenodo record 15108715.

Choose a segment to follow through the methods. The highest anomaly probabilities appear first. Selecting another example does not retrain or refit the model. The data overview and architecture map describe the whole program, so those views stay the same.

Example held-out test segment

1068

Selected segment 1068, channel CADC0888.

The segment ID identifies this example. The channel code identifies its telemetry source; it does not tell us which spacecraft subsystem caused an unusual reading.

A multilayer perceptron (MLP) is a neural network that combines numerical inputs to make a prediction. Here, it estimates whether a segment is anomalous. It can learn patterns involving several features together, such as a change in both the average value and the amount of variation.

Input: Segment 1068's 18 standardized numerical features.

Action: Pass values forward through the fitted 18-32-16-8-1 neural network.

Output: Anomaly probability 100.0% for this segment.

The network classifies numerical examples. The next stage compares its probability with a cutoff. Similarity search also uses the same list of scaled features.

The probability is a model estimate, not certainty, a physical diagnosis, or an explanation of cause.

Multilayer perceptron (MLP) probability

Supervised learning means learning from examples with known answers. Here, those answers are the dataset's reviewed normal or anomalous labels. Weights are numbers the network learns during training; they control how strongly one value affects the next layer. Hidden layers are the intermediate calculations between inputs and output.

This network is fully connected: each value in one layer connects to every unit in the next. The rectified linear unit (ReLU) keeps positive hidden-layer values and sets negative ones to zero. During training, backpropagation calculates how each weight contributes to error, and the optimizer uses that information to adjust the weights. Here, feedforward prediction means passing the selected example through the trained network without changing those weights.

Implemented shape: 18 inputs → 32 → 16 → 8 → 1 probability

The three hidden layers contain 32, 16, and 8 units, respectively.

Current example: Authentic OPS-SAT-AD v2, segment 1068, channel CADC0888. Channel is a source code, not an inferred subsystem name.

How to read the feature table

- Feature:** The exact OPS-SAT-AD source name for one calculated signal summary.
- Raw value:** The stored feature before this app scales it. It is already a summary of readings, rather than one original sensor reading.
- Standardized value:** The raw value measured in model-training standard deviations from the model-training mean.

The program calculates each standardized value as (raw value - model-training mean) / model-training standard deviation. A value near 0 is near the training average for that feature. Positive and negative values are above and below that average. Standardization puts differently scaled inputs on a comparable basis; it is not a percentage or a class label. A standard deviation describes the spread of values around their average. For example, a standardized value of +2 means two training standard deviations above that feature's average. A value of -2 is the same distance below it.

Feature	Raw value	Standardized value
n_peaks	1	-0.224
smooth10_n_peaks	1	-0.274
smooth20_n_peaks	1	-0.416
diff_peaks	1	-0.590
diff2_peaks	1	-0.748
diff_var	0.0998787	26.699
diff2_var	0.303399	33.335
gaps_squared	225	-0.485
len_weighted	50	-1.323
var_div_duration	0.00131344	5.753

▼ Feature glossary: what the 18 MLP inputs measure

Use the source name in parentheses to find the matching row in the feature table. Each explanation describes a numerical summary; a larger value alone does not identify a fault.

Elapsed duration (duration)

Seconds from the first timestamp to the last timestamp in the segment. Group: Time and size.

Reading count (len)

Number of telemetry readings in the segment. Group: Time and size.

Average value (mean)

Arithmetic average of the segment's telemetry values. Group: Signal distribution.

Variance (var)

Average squared spread of the telemetry values around their mean. Group: Signal distribution.

Standard deviation (std)

Square root of variance; spread expressed in the signal's original value scale. Group: Signal distribution.

Kurtosis (kurtosis)

Distribution-shape statistic that is sensitive to heavy tails and extreme values. Group: Signal distribution.

Skewness (skew)

Signed measure of asymmetry; positive and negative values show opposite tail directions. Group: Signal distribution.

Raw peak count (n_peaks)

Peaks whose prominence is at least 10% of the segment's value range. Group: Peaks.

10-reading smoothed peak count (smooth10_n_peaks)

Peak count after a 10-reading moving average reduces short fluctuations. Group: Smoothed peaks.

20-reading smoothed peak count (smooth20_n_peaks)

Peak count after a 20-reading moving average reduces short fluctuations. Group: Smoothed peaks.

First-difference peak count (diff_peaks)

Peak count in the changes between adjacent telemetry readings. Group: Changes.

Second-difference peak count (diff2_peaks)

Peak count in the changes between successive first differences. Group: Changes.

First-difference variance (diff_var)

Variance of the changes between adjacent readings. Group: Changes.

Second-difference variance (diff2_var)

Variance of how the change between adjacent readings changes again. Group: Changes.

Squared time-gap score (gaps_squared)

Sum of squared elapsed-second gaps; larger gaps contribute more to the score. Group: Gaps and scaled measures.

Sampling-weighted length (len_weighted)

The source sampling value multiplied by the number of readings. Group: Gaps and scaled measures.

Variance per duration (var_div_duration)

Signal variance divided by the segment duration. Group: Gaps and scaled measures.

Variance per reading (var_div_len)

Signal variance divided by the number of readings. Group: Gaps and scaled measures.

MLP anomaly probability

100.0%

The model returns a value from 0 to 1, displayed here as a percentage rounded to one decimal place. A displayed 100.0% can be a rounded value close to 1. It does not mean the prediction is guaranteed to be correct.



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Threshold evaluation

Choose one walkthrough stage

Data source and roles

MLP probability

Threshold and evaluation

Similarity search

Logic rules

A-star planning

Architecture map

Active data profile: Authentic OPS-SAT-AD v2. Historical OPS-SAT Anomaly Detection (OPS-SAT-AD) v2 data from Zenodo record 15108715.

Choose a segment to follow through the methods. The highest anomaly probabilities appear first. Selecting another example does not retrain or refit the model. The data overview and architecture map describe the whole program, so those views stay the same.

Example held-out test segment

1068

Selected segment 1068, channel CADC0888.

The segment ID identifies this example. The channel code identifies its telemetry source; it does not tell us which spacecraft subsystem caused an unusual reading.

This stage turns the MLP's probability into a review status, then compares all held-out predictions with their reviewed classes. The threshold is a cutoff chosen using the validation rows. A band around it marks close calls for review. Evaluation shows how often predictions agree with the recorded labels.

Input: One MLP probability, the chosen cutoff, and the uncertainty margin.

Action: Assign a review status and compare test predictions with reviewed labels.

Output: Flagged anomaly plus current evaluation metrics.

The status helps you decide which segments to inspect. The logic stage uses the classification and probability to form facts. The evaluation below measures the model across all test segments.

These values describe one fitted model on the selected local profile. They are not published benchmark results or operational spacecraft performance.

One classification calculation

Current example: Authentic OPS-SAT-AD v2, segment 1068, channel CADC0888.

Probability

100.0%

Threshold

71.0%

Uncertainty margin (plus or minus)

5.0%

AI screening status

Flagged anomaly

lower edge = 0.710 - 0.050 = 0.660

upper edge = 0.710 + 0.050 = 0.760

segment probability = 1.000 -> Flagged anomaly

The probability is above the 76.0% upper edge of the uncertainty band, so the AI flags this segment for review.

The margin is 5 percentage points on each side of the 71.0% threshold. Probabilities from 66.0% through 76.0%, including both edges, receive Uncertain review. This band is a review rule chosen for the app; it is not a statistical confidence interval.

Current held-out evaluation

These metrics use two predictions: anomalous at or above the threshold, and normal below it. A segment marked Uncertain review still receives one of those predictions here. That is why the confusion matrix can differ from the three screening-status counts on Analyze telemetry. Choosing another example does not change this evaluation; it covers the full test set for the current model.

Precision

94.3%

Recall

87.6%

F1 score

90.8%

Accuracy

96.2%

The evaluation denominator is 529 held-out segments: 416 reviewed normal and 113 reviewed anomalous. The reviewed classes are revealed for evaluation only after prediction.

What the evaluation metrics answer

- Precision:** Of all predicted anomalies, what fraction were reviewed anomalies? Higher precision means fewer false anomaly flags.
- Recall:** Of all reviewed anomalies, what fraction did the model detect? Higher recall means fewer missed anomalies.
- F1 score:** The harmonic mean of precision and recall, calculated as $2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$. It drops when either measure is low, so a high value needs both to be strong.
- Accuracy:** The fraction of all held-out predictions that were correct. If most examples are normal, many correct normal predictions can hide missed anomalies. Read accuracy together with recall.

How to read the confusion matrix

- Rows:** The reviewed normal or anomalous class for each held-out segment.
- Columns:** The normal or anomalous class predicted by the MLP and threshold.
- Reviewed normal / predicted normal:** Correctly classified normal segments.
- Reviewed normal / predicted anomalous:** False anomaly flags.
- Reviewed anomalous / predicted normal:** Missed anomalies.
- Reviewed anomalous / predicted anomalous:** Correctly detected anomalies.

	Predicted normal	Predicted anomalous
Actual normal	410	6
Actual anomalous	14	99

In this run, the model detected 99 of 113 reviewed anomalies and missed 14. It correctly classified 410 normal segments and raised 6 false anomaly flags.

For example, precision is 99 correct anomaly predictions / 105 total anomaly predictions = 94.3%.

Rows are reviewed classes and columns are model predictions. Diagonal cells are correct; off-diagonal cells are errors. The software test suite separately checks program behavior and does not improve these model-performance scores.

Threshold Similar search

Choose one walkthrough stage

Data source and roles

MLP probability

Threshold and evaluation

Similarity search

Logic rules

A-star planning

Architecture map

Active data profile: Authentic OPS-SAT-AD v2. Historical OPS-SAT Anomaly Detection (OPS-SAT-AD) v2 data from Zenodo record 15108715.

Choose a segment to follow through the methods. The highest anomaly probabilities appear first. Selecting another example does not retrain or refit the model. The data overview and architecture map describe the whole program, so those views stay the same.

Example held-out test segment

1068

Selected segment 1068, channel CADC0888.

The segment ID identifies this example. The channel code identifies its telemetry source; it does not tell us which spacecraft subsystem caused an unusual reading.

Similarity search is a historical comparison tool. It finds the recorded segments whose 18 feature values most closely resemble the selected example. These are called nearest neighbors. A reviewer can compare the selected segment with known normal and anomalous examples instead of relying on the MLP probability alone.

Input: Segment 1068's feature vector: its ordered list of 18 scaled values.

Action: Measure Euclidean distance to historical examples, searching the reviewed normal and anomalous groups separately.

Output: Separate nearest normal and anomalous historical cases.

These comparisons add context to the MLP prediction. The search reuses the standardized feature vector from the MLP stage. Its historical evidence can become explicit facts for the logic stage.

Numerical similarity does not transfer a neighbor's label to the selected segment and does not prove that two segments have the same physical cause.

Current example: Authentic OPS-SAT-AD v2, segment 1068, channel CADC0888. Historical references include both the model-training and threshold-validation rows. Held-out test rows are excluded. The search groups references by reviewed class, not by channel, so a neighbor may be from another channel.

Euclidean distance combines the differences across all 18 scaled features into one number. A smaller value means the two examples are more alike. A neighbor's reviewed label supplies context; it never becomes the selected segment's prediction.

How to read each similarity table

- Rank:** Position within that reviewed class; rank 1 has the smallest distance.
- Reviewed case:** The segment identifier of the historical comparison record.
- Reviewed label:** The historical normal or anomalous class assigned by reviewers.
- Euclidean distance:** The square root of the summed squared gaps across all 18 standardized features; smaller means closer in this feature space.
- Largest standardized feature gaps:** The inputs on which the selected segment and neighbor differ most, measured in training-standard-deviation units. These gaps explain how the two examples differ; they are not scores of feature importance in the MLP.

The normal and anomalous lists are separate so a larger class cannot hide the nearest examples from the other class. Distance is a comparison value, not a probability.

Nearest reviewed normal cases

Rank	Reviewed case	Reviewed label	Euclidean distance	Largest standardized feature gaps
1	234	normal	41.507	diff2_var (33.053), diff_var (24.038), skew (4.172)
2	1076	normal	41.802	diff2_var (33.105), diff_var (24.538), var_div_len (4.449)
3	223	normal	41.919	diff2_var (33.268), diff_var (24.615), skew (4.023)

- Rank 1 is segment 234, reviewed as normal, at distance 41.507. Its largest feature gaps are diff2_var (33.053), diff_var (24.038), skew (4.172).
- Rank 2 is segment 1076, reviewed as normal, at distance 41.802. Its largest feature gaps are diff2_var (33.105), diff_var (24.538), var_div_len (4.449).
- Rank 3 is segment 223, reviewed as normal, at distance 41.919. Its largest feature gaps are diff2_var (33.268), diff_var (24.615), skew (4.023).

Nearest reviewed anomalous cases

Rank	Reviewed case	Reviewed label	Euclidean distance	Largest standardized feature gaps
1	1726	anomalous	11.007	diff2_var (7.317), skew (4.220), diff_var (3.996)
2	1727	anomalous	11.608	diff_var (6.671), diff2_var (6.365), skew (4.323)
3	1719	anomalous	27.973	diff2_var (22.843), diff_var (14.535), skew (4.264)

- Rank 1 is segment 1726, reviewed as anomalous, at distance 11.007. Its largest feature gaps are diff2_var (7.317), skew (4.220), diff_var (3.996).
- Rank 2 is segment 1727, reviewed as anomalous, at distance 11.608. Its largest feature gaps are diff_var (6.671), diff2_var (6.365), skew (4.323).
- Rank 3 is segment 1719, reviewed as anomalous, at distance 27.973. Its largest feature gaps are diff2_var (22.843), diff_var (14.535), skew (4.264).

Distances and largest feature gaps come from the current fitted backend. They are comparison evidence, not proof of the same cause.

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Logic rules

Choose one walkthrough stage

- Data source and roles
- MLP probability
- Threshold and evaluation
- Similarity search
- Logic rules
- A-star planning
- Architecture map

Active data profile: Authentic OPS-SAT-AD v2. Historical OPS-SAT Anomaly Detection (OPS-SAT-AD) v2 data from Zenodo record 15108715.

Choose a segment to follow through the methods. The highest anomaly probabilities appear first. Selecting another example does not retrain or refit the model. The data overview and architecture map describe the whole program, so those views stay the same.

Example held-out test segment

1068

Selected segment 1068, channel CADC0888.

The segment ID identifies this example. The channel code identifies its telemetry source; it does not tell us which spacecraft subsystem caused an unusual reading.

An expert system applies written rules to facts. Here, the rules connect a segment's classification and historical comparisons to checks a person can review. Each rule that fires records its supporting facts and conclusion, so you can follow how the program reached that result.

Input: Calculated results, reviewed historical labels, and configured assumptions.

Action: Match the conditions in each if/then rule to the available facts.

Output: Rule-derived conclusions, review reasons, and simulated hypotheses.

The rules explain which conditions call for further review. The rule engine receives the classification and historical comparisons. Any required checks then pass to A-star planning.

The engine can use only facts and rules encoded by this program. It does not learn the MLP probability or confirm a physical fault.

Current example: Authentic OPS-SAT-AD v2, segment 1068, channel CADC0888.

A probability is a number. A fact is a statement the program currently treats as true, such as `Uncertain(segment_id)`. This is Boolean logic: a rule condition either matches a known fact or does not. A missing fact does not establish that its opposite is true.

Facts available to the expert system

Starting facts combine calculated results, reviewed labels, and teaching assumptions. Critical and FaultMapping come from configured assumptions, including when you select authentic data. They do not identify a verified critical channel or spacecraft fault.

The rule engine checks uncertainty first. For other segments, it uses a separate 80% cutoff to add Anomalous. A Flagged anomaly screening status below that cutoff can therefore appear as insufficientEvidence here. Deviates means at least one feature is 2 or more standard deviations from the reviewed normal model-training average. This normal-only comparison differs from the MLP scaler, which uses both training classes.

How to read the fact table

- Fact role:** Starting fact means it was supplied to the rules. Derived by a rule means the program added it after matching rule conditions.
- Fact:** A named statement, or predicate, such as `Uncertain(segment)`. The text in parentheses identifies the segment or other item the statement describes.

Fact role	Fact
Starting fact	Anomalous(1068)
Starting fact	Deviates(1068, CADC0888)
Starting fact	FaultMapping(CADC0888, simulated_attitude_subsystem)
Starting fact	HistoricalSupports(1068, anomalous)
Starting fact	ReviewedLabel(1076, normal)
Starting fact	ReviewedLabel(1719, anomalous)
Starting fact	ReviewedLabel(1726, anomalous)
Starting fact	ReviewedLabel(1727, anomalous)
Starting fact	ReviewedLabel(223, normal)
Starting fact	ReviewedLabel(234, normal)

Rules that fired

A Horn clause is an if/then rule with one conclusion. Its premises are the conditions that must all match before it fires. For example, `Uncertain(?s) → ManualReviewRequired(?s)` means: if segment ?s is uncertain, require manual review of that segment. The ?s is a placeholder for a segment ID; the arrow means 'then'. This example explains the rule, whether or not it fires for your selection.

Unification matches that placeholder to an actual ID and keeps the match consistent through the rule. The recorded match is a variable binding. Forward chaining repeats this process, using newly added facts to check more rules until no new facts are added. A trace is the record of one rule application. Any resulting fault name remains a simulated hypothesis to investigate.

How to read a fired-rule trace

- Rule:** The stable identifier of the Horn clause that matched.
- Matched facts:** Known facts that satisfied every premise of that rule.
- Variable bindings:** The constants substituted for placeholders such as ?s for segment.
- Conclusion:** The new fact added after all premises matched.
- Why it fired:** A plain-language explanation of the rule's intended reasoning.
- Source status:** Whether the rule pattern is source-derived or a simulated educational rule; it is not a confidence score.

No Horn rule fired for this selected segment because its current facts did not satisfy every premise of any rule. This is a valid no-match result, not an engine failure.

The probability remains the MLP output; rules operate on explicit facts after that calculation.

A-star planning

Choose one walkthrough stage

Data source and roles	MLP probability	Threshold and evaluation	Similarity search	Logic rules	A-star planning	Architecture map
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Active data profile: Authentic OPS-SAT-AD v2. Historical OPS-SAT Anomaly Detection (OPS-SAT-AD) v2 data from Zenodo record 15108715.

Choose a segment to follow through the methods. The highest anomaly probabilities appear first. Selecting another example does not retrain or refit the model. The data overview and architecture map describe the whole program, so those views stay the same.

Example held-out test segment

1068

Selected segment 1068, channel CADCO888.

The segment ID identifies this example. The channel code identifies its telemetry source; it does not tell us which spacecraft subsystem caused an unusual reading.

A-star (A*) is a graph-search algorithm that chooses a sequence of steps. Each node records which checks are complete and which remain. Each connection, or edge, represents an allowed simulated review action. It compares the cost so far with an estimate of the work left to find a low-cost sequence that completes the required checks.

Input: A planning state, required checks, allowed actions, costs, and a goal.

Action: Use A-star to compare candidate states with $f(n) = g(n) + h(n)$.

Output: Least-cost simulated advisory investigation sequence found.

A-star orders the review checks; it does not detect anomalies. It receives required checks derived from the classification and logic stages, then returns an advisory sequence for a human reviewer.

The costs are coursework values and the actions are simulated. The plan does not execute a command, inspect hardware, or prove that a check occurred.

Current example: Authentic OPS-SAT-AD v2, segment 1068, channel CADCO888. Every action and cost below is simulated.

In $f(n) = g(n) + h(n)$, n is one planning state. $g(n)$ is the cost of the simulated steps taken so far. $h(n)$, called the heuristic, estimates the least cost still needed. A-star considers smaller totals, $f(n)$, first. The costs represent relative review effort chosen for this coursework example; they are not minutes or money.

Starting route: anomalous

Goal: complete all mandatory simulated checks: historical_cases_reviewed, probability_reviewed

Allowed simulated actions

How to read the action table

- Action:** The stable identifier for one allowed simulated review step.
- Description:** What evidence the simulated step would review.
- Preconditions:** Facts that must already be true before the action is allowed.
- Effects:** Checks marked complete after the simulated action is applied.
- Cost:** A nonnegative coursework value used to compare possible sequences.
- Selected:** Whether the final A-star path includes that action.

Action	Description	Preconditions	Effects	Cost	Selecte
ACTION-001-REVIEW-PROBABILITY	Review the model probability and validation-selected threshold.	None	probability_reviewed	1.0	Yes
ACTION-002-INSPECT-HISTORICAL-CASES	Inspect class-aware normal and anomalous historical cases.	probability_reviewed	historical_cases_reviewed	1.0	Yes
ACTION-003-INSPECT-CRITICAL-DEVIATION	Inspect the critical channel deviation and its contributing features.	historical_cases_reviewed	critical_deviation_reviewed	2.0	No
ACTION-004-REVIEW-FAULT-MAPPING	Review the simulated channel-to-fault hypothesis mapping.	critical_deviation_reviewed	fault_mapping_reviewed	2.0	No
ACTION-005-ESTIMATE-MANUAL-REVIEW	Estimate the evidence necessary for final human interpretation	historical_cases_reviewed	manual_review_estimated	1.0	No

Chosen path and simulated costs

This table follows only the chosen sequence, starting at step 0. A-star may examine other states while searching; those alternatives are included in the expanded-node count but are not listed here.

How to read the selected path

- Step:** The position in the chosen sequence; step 0 is the starting state.
- Selected action:** The action that produced the state shown on that row.
- Candidate state:** Which mandatory checks are complete and which still remain after the action.
- $g(n)$ accumulated:** The simulated action cost already spent to reach this state.
- $h(n)$ remaining estimate:** A conservative estimate of the least cost still needed to finish required checks.
- $f(n)$ total estimate:** The priority value $g(n) + h(n)$; A-star considers lower values first.

Step	Selected action	Candidate state	$g(n)$ accumulated	$h(n)$ remaining estimate	$f(n)$ total es
0	Start state	completed: none; remaining: historical_cases_reviewed, probability_reviewed	0.0	2.0	
1	ACTION-001-REVIEW-PROBABILITY	completed: probability_reviewed; remaining: historical_cases_reviewed	1.0	1.0	
2	ACTION-002-INSPECT-HISTORICAL-CASES	completed: historical_cases_reviewed, probability_reviewed; remaining: none	2.0	0.0	

At step 0, no action has been taken, so $g(n) = 0.0$. The estimated work left is $h(n) = 2.0$, giving $f(n) = 2.0$. As checks are marked complete in the simulation, $g(n)$ increases and the remaining estimate falls toward zero.

- Step 0: Start state. State: completed: none; remaining: historical_cases_reviewed, probability_reviewed. Cost so far $g(n) = 0.0$, $h(n) = 2.0$, $f(n) = 2.0$.
- Step 1: ACTION-001-REVIEW-PROBABILITY. State: completed: probability_reviewed; remaining: historical_cases_reviewed. Cost so far $g(n) = 1.0$, $h(n) = 1.0$, $f(n) = 2.0$.
- Step 2: ACTION-002-INSPECT-HISTORICAL-CASES. State: completed: historical_cases_reviewed, probability_reviewed; remaining: none. Cost so far $g(n) = 2.0$, $h(n) = 0.0$, $f(n) = 2.0$.

Total simulated cost 2.0	Expanded nodes 2	Goal satisfied Yes
-----------------------------	---------------------	-----------------------

Total simulated cost is the final $g(n)$. Expanded nodes counts the planning states A-star examined while finding the path. Goal satisfied means every mandatory simulated check is complete in the final state.

The sequence is advisory coursework output. It does not execute a command, diagnose hardware, or show that a physical check occurred.

Architecture map

Choose one walkthrough stage

Data source and roles

MLP probability

Threshold and evaluation

Similarity search

Logic rules

A-star planning

Architecture map

Active data profile: Authentic OPS-SAT-AD v2. Historical OPS-SAT Anomaly Detection (OPS-SAT-AD) v2 data from Zenodo record 15108715.

Choose a segment to follow through the methods. The highest anomaly probabilities appear first. Selecting another example does not retrain or refit the model. The data overview and architecture map describe the whole program, so those views stay the same.

Example held-out test segment

1068

Selected segment 1068, channel CADC0888.

The segment ID identifies this example. The channel code identifies its telemetry source; it does not tell us which spacecraft subsystem caused an unusual reading.

This map shows which source file handles each part of the app and which course module introduced the method. It lets a learner trace a screen value back to the code that supplies it and see how earlier course modules contribute to the integrated Module 8 program.

Input: The program stages and the results displayed on these pages.

Action: Connect each stage to its source file and course milestone.

Output: A guide to where each result comes from.

The backend is the Python code that calculates results. The frontend is the interface you use to choose examples and read those results. The presentation adapter converts the calculated results into tables and charts.

This is a guide to responsibilities, not a runtime call trace: it does not record which functions ran when you selected an example.

Current profile context: Authentic OPS-SAT-AD v2.

How to read the architecture map

- **Stage:** One problem, data, AI-method, coordination, or presentation responsibility.
- **Input:** The information that responsibility receives.
- **Output:** The record or decision support that responsibility produces.
- **Current source:** The current document or Python module that supplies the behavior.
- **Layer:** Backend calculates results, frontend presents them, and design evidence records the problem and intended human decision.
- **Course lineage:** The earlier course module or Module 8 integration work demonstrated by the stage.

Stage	Input	Output	Current source	
Problem and human decision	CubeSat telemetry use case	Human-review objective	Milestone Module 2 and README.md	L
Data validation and roles	Local OPS-SAT-AD comma-separated value (CSV) files	Checked dataset with training, validation, and test groups	cubesat_data.py	E
MLP fitting and classification	18 numerical features and labels	Probability, threshold, and metrics	cubesat_model.py	E
Historical similarity	Selected feature values and historical training examples	Similar normal and anomalous examples in separate groups	cubesat_similarity.py	E
Logic and expert system	Numerical and similarity evidence	Facts, traces, and hypotheses	cubesat_logic.py	E
A-star planning	Planning state, actions, costs, and goal	Simulated advisory action sequence	cubesat_planner.py	E
Analysis coordination	Selected segment	CompleteAnalysisResult (the combined results for one segment)	cubesat_telemetry_ai.py	E
Presentation adapter	Validated backend objects	Tables, chart, and walkthrough records	cubesat_frontend.py	F
Interactive pages	Presentation records and user selections	Screening and learner walkthrough	streamlit_app.py and app_pages/	F

- **Problem and human decision.** The Module 2 milestone and README.md describe the CubeSat use case and the decision a person needs to make.
- **Data validation and roles.** `cubesat_data.py` checks the files and separates training, validation, and test rows. This builds on Modules 2 and 5.
- **MLP fitting and classification.** `cubesat_model.py` learns from the 18 features and reviewed labels, then supplies probabilities, a threshold, and evaluation metrics. The neural-network methods connect to Modules 3 and 5.
- **Historical similarity.** `cubesat_similarity.py` finds nearby normal and anomalous examples using the selected feature values. This applies search concepts from Module 4.
- **Logic and expert system.** `cubesat_logic.py` turns results and teaching assumptions into facts, then applies rules and records their conclusions. This uses Module 6 knowledge-representation concepts.
- **A-star planning.** `cubesat_planner.py` uses states, allowed actions, and costs to choose simulated review steps. A-star also connects to Module 4.
- **Analysis coordination.** `cubesat_telemetry_ai.py` calls these methods for one segment and collects their outputs in `CompleteAnalysisResult`. This is part of the Module 8 integration.
- **Presentation adapter.** `cubesat_frontend.py` turns those results into tables, charts, and walkthrough examples for the Module 8 interface.
- **Interactive pages.** `streamlit_app.py` and `app_pages/` display the results and handle your selections. They form the frontend added for Module 8.

The milestone PDFs establish course lineage. The current Python modules determine the behavior shown in this app. Analyze telemetry is the screening workflow; this page is the learner walkthrough.

